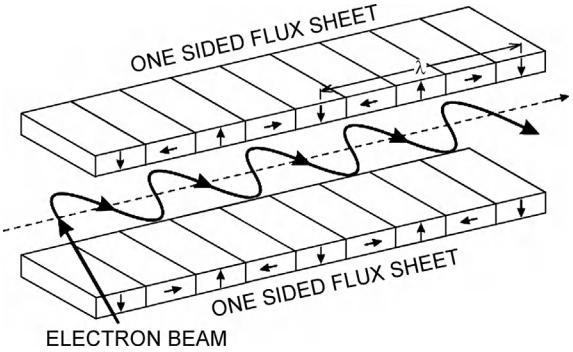


Figure 16.12 Principle of undulators and wigglers for production of intense synchrotron radiation. The X-rays are emitted in the direction shown by the dashed line with an arrow at the right.

shown schematically in Figure 16.12. The direction changes will increase the intensity of the x-ray beam due to the large number of bendings. Syncrotron light sources are used for different kinds of spectroscopy such as, e.g., EXAFS (Extended X-ray Absorption Fine Structure) which has grown considerably in popularity with chemists in the last decade due to the ability to directly study complexation in solution and solids.

16.10. EXERCISES

16.1. In a small linear accelerator containing 30 stages, He^{2+} ions are accelerated by a 150 kV, 100 MHz RF source. The ions are used to bombard a metal target to induce a specific reaction. (a) What is the proper length of the last drift tube? (b) What is the maximum projectile energy achieved? (c) What is the heaviest target in which a nuclear transformation can be induced (no tunnelling)?

16.2. Assume a linear accelerator is built in three sections, each with 30 stages of 100 kV. Between which sections should a thin carbon stripper foil be placed in order to achieve the highest final energy, when accelerating ^{84}Kr ions from an ion source emitting low energy $^{84}Kr^{4+}$ ions? Consider the nearest integer charge after stripping.

16.3. In order to propel a space vehicle to high velocities after its exit from the earth's gravitational pull, "ion rockets" might be used. These can be considered as simple accelerators for charged particles. The electroneutrality of the vehicle is conserved through emission of electrons from a hot filament. The gain in linear velocity Δv is given by the "rocket formula":

$$\Delta v = v_e \ln(m_{Ro}/m_R)$$

where v_e is the exhaust velocity and m_{Ro} and m_R are the initial and final mass of the vehicle respectively. (a) Calculate the propelling power for a rocket, which emits 10 keV protons at a current of 1 A. (b) What is the final velocity gain of such vehicle with 2000 kg initial mass after 1 year's operating time? (c) For the same net available power and operating time, as in (b), calculate the final velocity gain of a 2000 kg vehicle which emits 10 keV Cs^+ ions at a current of 1 A.

16.4. The TD reaction is used to produce 14 MeV neutrons, which are considered to be emitted isotropically from the target. What is the fast neutron "flux" at 5 cm from the target when the ion current is 0.2 mA and the acceleration voltage is 300 kV? Use Figure 16.11.

16.5. Protons are accelerated to 12 GeV in a synchrotron in which the bending magnets have a maximum field strength of 14.3 T. What is the radius of curvature of the proton orbit?

16.6. In a linear accelerator for protons the first drift tube is 1 cm long and the accelerating potential is 25 kV. (a) The speed of light is not approached in the first drift tube. How long should the second and third drift tubes be? (b) What should the full-wave frequency of acceleration be?

16.7. Calculate the maximum energy (a) for protons, deuterons, and helium ions in a cyclotron, whose maximum orbit diameter is 1.25 m and whose frequency is 12 MHz. (b) What magnetic field strength would be required in each case?

16.8. A thin Au target is irradiated with a beam of 800 MeV $^{16}O^{2+}$ ions during 2 hours. After the irradiaten a faraday cup (with current integrator) behind the target shows a total accumulated charge of 92.3 μC. What was the average beam intensity (oxygen ions per second)? Consider the charge-stripping effect of the target.

16.9. In a VdG accelerator a 100 μA beam of He^{2+} ions is accelerated to an energy of 5 MeV before striking a target. How many grams of radium are required to provide the same number of α-particles?

16.10. A cyclotron can accelerate $^{4}He^{2+}$-ions to 35 MeV. (a) What is its *K*-value? To what energy would it accelerate (b) $^{16}O^{6+}$ and (c) $^{18}O^{8+}$ ions?

16.11. LITERATURE

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